

Circles, Cyclicity & Tangency

Diagnostic and benchmark teacher companion

STRUCTURE → JUSTIFY → LOCAL RELATION → ROUTE → CHECK

Teacher-only companion · independent answer and misconception diagnostics

Teacher Diagnostic Key — Circles, Cyclicity and Tangency

Use this companion to diagnose **representation and theorem-selection errors**, not only final answers.

Core diagnostic codes:

- CYCLICITY_ASSUMED_FROM_PICTURE
 - GENERIC_QUAD_RULE_USED
 - TANGENCY_NOT_PROVED
 - TANGENT_CHORD_CONFUSED
 - POWER_POINT_MISMATCH
 - WRONG_PRODUCT_SEGMENTS
 - SYNTHETIC_CHAIN_NOT_RECOGNIZED
 - COORDINATE_OVERKILL
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Recognition and First-Line Lab

1. **C.** Without cyclicity, only the generic quadrilateral angle sum is safe.
2. **A.** Same chord AB, same segment $\rightarrow$ equal inscribed angles.
3. **B.** A diameter subtends a right angle at the circumference.
4. **B.** Radius to the point of tangency is perpendicular to the tangent.
5. **A.** Tangents from the same external point to the same circle are equal.
6. **B.** Tangent + chord is the direct recognition cue for alternate segment.
7. **B.** The power point is P; pair the two pieces on each chord through P.
8. **B.** The line of centres is perpendicular to the common chord.
9. First line: $\angle A + \angle C = 180^\circ$, hence $\angle C = 68^\circ$.
10. Conclude ABCD is cyclic by the converse of the opposite-supplementary-angle theorem.
11. First mark OT perpendicular to the tangent at T; only then translate the tangent-chord angle via alternate segment if useful.
12. $PT^2 = PA \cdot PB$.
13. $PA \cdot PB = PC \cdot PD$.
14. Test coordinates when the right/square metric structure gives a low-variable placement and the circle condition becomes one short equation.
15. Missing target: prove cyclicity, for example by a supplementary opposite-angle pair or a valid equal-subtended-angle converse.
16. The line of centres is perpendicular to AB; the other stated line is also perpendicular to AB, so the two lines are parallel.

Diagnostic emphasis:

- 1,9,10,15 $\rightarrow$ CYCLICITY_ASSUMED_FROM_PICTURE or GENERIC_QUAD_RULE_USED;
 - 4,6,11 $\rightarrow$ TANGENCY_NOT_PROVED / TANGENT_CHORD_CONFUSED;
 - 7,12,13 $\rightarrow$ POWER_POINT_MISMATCH / WRONG_PRODUCT_SEGMENTS;
 - 14,16 $\rightarrow$ route-selection diagnosis.
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Practice and Transfer Bank

1. 64° .
2. 76° .
3. 90° .
4. 13.
5. 37° .
6. The quadrilateral is cyclic because $72^\circ + 108^\circ = 180^\circ$.
7. $\angle ACB = 90^\circ$; with hypotenuse $AB = 10$ and leg $AC = 6$, $BC = 8$.
8. $12^2 = 8 \cdot PB$, so $PB = 18$.
9. $3 \cdot 12 = 4 \cdot PD$, so $PD = 9$.
10. $5 \cdot 20 = 4 \cdot PD$, so $PD = 25$.
11. 48° by alternate segment.
12. Equal tangents give $PB = 10$; perimeter $10 + 10 + 12 = 32$.

13. Join centre O to contact points A, B . Triangles OAP and OBP are right triangles with $OA = OB$ and common hypotenuse OP ; hence congruent, so $PA = PB$.
14. $\angle ABD$ and $\angle ACD$ both subtend segment AD . Equality is the converse same-segment signal; justify A, B, C, D concyclic before using cyclic relations.
15. $O1O2$ is perpendicular to common chord AB . The additional line is also perpendicular to AB , hence it is parallel to $O1O2$.
16. The rectangle supplies natural orthogonal coordinates and the equal lengths give low-degree equations. Concyclicity must still be imposed explicitly; it is the decisive extra constraint.
17. 25. Independent historical reconstruction: symmetry-axis coordinate placement, circumcentre on the axis, horizontal chord through the midpoint.
18. 29. Independent historical reconstruction gives square side $2/5$, area $4/25$, hence $4 + 25 = 29$.
19. 03. The canonical non-degenerate branch gives $(AB/BC)^2 = 2/1$.
20. 10. Common-chord/line-of-centres structure plus internal tangency yields radius sum 10.
21. 03. Independent circumcentre calculation gives $(OP/OA)^2 = 1/2$.
22. Power is $PA \cdot PB = 4 \cdot 9 = 36$. Thus $6 \cdot PD = 36$, so $PD = 6$; also $PT^2 = 36$, hence $PT = 6$.

Historical answers 25, 29, 03, 10, 03 were independently re-derived in Authoring/Independent_Math_and_Sou before this key was written.

Mixed Mastery Test

1. 73° .
2. 112° .
3. The quadrilateral is cyclic because $95^\circ + 85^\circ = 180^\circ$.
4. $15^2 = 9 \cdot PB$, so $PB = 25$.
5. $6 \cdot 8 = 4 \cdot PD$, so $PD = 12$.
6. $3 \cdot 18 = 6 \cdot PD$, so $PD = 9$.
7. 41° .
8. Equal tangents give $PA = PB = 17$. With included angle 60° , the cosine rule gives $AB^2 = 17^2 + 17^2 - 2 \cdot 17 \cdot 17 \cdot \cos 60^\circ = 17^2$, hence $AB = 17$.
9. Since $O1A = O1B$, centre $O1$ lies on the perpendicular bisector of chord AB . Likewise $O2A = O2B$, so $O2$ lies on the same perpendicular bisector. Therefore $O1O2$ is perpendicular to AB .
10. Distance from centre $(0, 0)$ to chord line $y = 3$ is 3. Half-chord length is $\sqrt{25 - 9} = 4$; full chord length 8. Coordinates are natural because the circle and horizontal chord are already in coordinate form.
11. The move is invalid because tangency is neither stated nor proved; visual contact is not a hypothesis. It is sufficient to know l is tangent at T , or to prove T lies on the circle and OT is perpendicular to l , which establishes tangency.
12. Alternate segment gives $\angle ACB = 52^\circ$. If D lies in the same segment as C and also subtends chord AB , then $\angle ADB = \angle ACB = 52^\circ$.

Failure interpretation

Cyclicity errors

If a learner uses supplementary opposite angles without establishing cyclicity, require them to rewrite the solution in two columns:

- what is known before cyclicity;
- what becomes legal only after cyclicity.

Tangency errors

If a learner uses a tangent theorem from appearance, require the exact tangent hypothesis or a radius-perpendicular proof before any angle work.

Power errors

If a learner writes the wrong product, have them circle the power point and trace each entire line through it. The two factors in each product must belong to the same line from that point.

Route-selection errors

If the learner uses coordinates for a one-step angle theorem, ask for a synthetic first move only. If the learner persists with a long synthetic chain in a symmetric metric rectangle/right-triangle setup, ask them to design a coordinate placement before solving.

Static evidence boundary

All deterministic learner-item answers/routes in this key were independently recomputed from the authored statements. This is static authoring evidence only. Classroom timing/readability, longitudinal retention, psychometric calibration, qualification/pass-mark calibration and publication approval remain NOT_RUN.

Teacher Benchmark Assimilation Key — Circles, Cyclicity and Tangency

This key evaluates whether the learner can **earn and select** a circle relation rather than merely recall theorem names.

A. RECONNECT

1. Check that cyclicity is stated or proved before using opposite supplementary angles.
2. Test equality: inscribed angles subtending the same chord from the same segment are equal.
3. Radius to the point of tangency is perpendicular to the tangent.
4. The tangent segments must start from the same external point and touch the same circle.
5. The shared chord links the tangent-chord angle to the inscribed angle in the alternate segment.
6. Identify the point whose power with respect to the fixed circle is being computed.
7. The line joining the two centres is perpendicular to the common chord.
8. Test coordinates when symmetry/right angles/rectangles/squares and numerical metric data make a low-variable coordinate model cheaper than a long synthetic chain.

Suggested diagnostic codes:

- 1-2: CYCLICITY_ASSUMED_FROM_PICTURE / same-chord recognition weakness;
 - 3-5: TANGENCY_NOT_PROVED / TANGENT_CHORD_CONFUSED;
 - 6: POWER_POINT_MISMATCH;
 - 7: intersecting-circle recognition weakness;
 - 8: COORDINATE_OVERKILL or synthetic overcommitment.
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B. Error laboratory

Error 1

First invalid move: treating visual concyclicity as a hypothesis.

Repair: prove cyclicity, for example from supplementary opposite angles or an appropriate equal-angle converse, or cite the explicit concyclicity statement.

Error 2

First invalid move: reading tangency from the sketch.

Repair: require explicit tangency or prove that the line is perpendicular to the radius at a point on the circle.

Error 3

First invalid move: pairing segments by size rather than by complete line through the power point.

Repair: for chords through P, use $PA \cdot PB = PC \cdot PD$ with A, B on one line and C, D on the other.

Error 4

First invalid move: using outside piece $\times$ inside piece.

Repair: for an external secant P-A-B, use $PA \cdot PB$, where PB is the whole distance from P to the far intersection.

Error 5

First inefficient move: theorem catalogue search before identifying structure.

Repair: write the structure first — same chord, cyclicity, tangent, power point, common chord — then choose the smallest theorem that uses it.

Error 6

First conceptual error: treating “circle problem” as synonymous with “synthetic only.”

Repair: preserve the circle condition explicitly, then compare synthetic and coordinate cost.

C. ADOPT — expected first two lines

1. ABCD cyclic $\rightarrow$ opposite angle to 117° equals 63° .
2. $83^\circ + 97^\circ = 180^\circ \rightarrow$ conclude the quadrilateral is cyclic.
3. Mark OT perpendicular to the tangent at T; then identify the chord seen by the tangent-chord angle.
4. Mark the two tangent lengths equal and connect the centre to the contact points if a proof is required.
5. Identify P as the power point and write $PA \cdot PB = PC \cdot PD$.
6. Identify P as the power point and write $PT^2 = PA \cdot PB$.
7. State that the line of centres is perpendicular to common chord AB.
8. Place the base symmetrically on an axis and put the apex/centre on the symmetry axis; then use the circle/chord distance relation.

D. TRANSFER — route expectations

1. The 180° opposite-angle sum proves cyclicity. After that upgrade, opposite supplementary and same-chord circle relations become legal.
2. Identify the common external point. If the road segment is tangent PT and another path is secant PAB, write $PT^2 = PA \cdot PB$.
3. Each centre is equidistant from A, B, so both lie on the perpendicular bisector of AB; hence the line joining the centres is perpendicular to AB.
4. The rectangle alone does not determine the metric ratio. Concurrency must be converted into an algebraic circle relation; it is the extra constraint that removes the remaining degree of freedom.
5. If the target angle and the tangent-chord angle see the same chord, alternate segment is one step and is cheaper. Otherwise continue with ordinary angle closure or another justified circle relation.
6. Accept any coherent coordinate-friendly example, such as a symmetric isosceles triangle with a circumcircle and horizontal chord. The circle condition must still appear explicitly as a radius/distance/circle equation or an equivalent chord-distance relation.

E. Six-question assimilation rubric

A strong response should contain all six components:

1. **Notice:** identifies the actual circle structure rather than naming a chapter.
2. **Legality:** states the hypothesis/proof that earns the relation.
3. **Recognition cue:** gives a transferable clue such as same chord, tangent + chord, power point, or common chord.
4. **Contrast:** names a similar-looking case where the theorem would be illegal or inefficient.
5. **First move:** writes two mathematically useful lines without solution scaffolding.
6. **Disguise:** changes the surface while preserving the same invariant/structure and outlines a valid route.

Do not award full assimilation credit to a learner who can solve one familiar numeric item but cannot explain why cyclicity/tangency/power is legal or cannot start a changed-surface version.

Static evidence boundary

All deterministic benchmark-lab routes were independently checked against the authored prompts. Classroom timing/readability, retention, psychometric calibration, qualification/pass-mark calibration and publication approval remain NOT_RUN.